

Toward a Theory of the Question of Existence (TQE): A Conceptual Framework for Law Emergence via Energy-Information Coupling

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Abstract

Why do stable, complexity-permitting physical laws exist at all? This question lies at the foundation of cosmology and fundamental physics. I propose a conceptual framework—the Theory of the Question of Existence (TQE)—where physical laws emerge dynamically from the coupling of vacuum energy fluctuations with an information-theoretic orientation parameter. The framework postulates that the universe originates from a quantum fluctuation in a pre-law state devoid of classical space-time, where a probabilistic modulation rule governs the selection of law-compatible configurations. At the mathematical core lies the equation $P'(\psi) = P(\psi) \cdot f(E, I)$, where $P(\psi)$ represents the probability distribution over possible law-states ψ , E denotes vacuum energy fluctuations, and $I \in [0, 1]$ is an information parameter measuring orientation toward complexity. **I acknowledge that the fundamental definition of I remains an open theoretical question, and this manuscript presents TQE as a conceptual framework rather than a complete theory.** Despite this fundamental uncertainty, the coupling function $f(E, I)$ creates a narrow “Goldilocks window” in parameter space, allowing only universes with the correct energy-information balance to stabilize their laws through a lock-in mechanism. This reframes the fine-tuning problem: rather than being pre-ordained, physical laws crystallize from the fundamental interaction of energy and information during cosmogenesis. The

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framework’s value lies in its conceptual architecture and testable consequences: it yields falsifiable predictions regarding statistical patterns in cosmological observations, even without a complete definition of I . TQE thus offers a theoretical framework for law emergence that moves beyond anthropic reasoning toward testable physics, while explicitly recognizing its current limitations.

1 Introduction

1.1 The Problem of Fine-Tuning

One of the most profound questions in modern cosmology is why the physical constants and laws of nature appear to be exquisitely fine-tuned to permit the emergence of complexity, structure, and ultimately observers. The cosmological constant Λ , the strength of fundamental forces, and the masses of elementary particles all exhibit values that, if slightly different, would render the universe incapable of supporting complex structures [2, 8]. This fine-tuning problem has generated numerous explanatory frameworks, each addressing different aspects of the puzzle.

The anthropic principle provides a retrospective justification: the laws are as they are because we exist to observe them [1, 3]. While logically consistent, this approach lacks predictive power and mechanistic content. The multiverse hypothesis and string landscape scenario propose that the universe is but one among an infinite or vast ensemble, selected by anthropic filtering [10, 12]. However, these frameworks often rely on assumptions that are difficult to verify empirically. Inflationary cosmology addresses the fine-tuning of initial conditions and expansion dynamics [5], yet it does not explain why particular configurations of physical laws are preferentially realized.

A deeper question underlies these discussions: *Are physical laws pre-existing, fundamental axioms of nature, or do they emerge dynamically during cosmogenesis?* The standard view in physics treats laws as immutable foundations that have always existed in their current form. However, this view raises the question of why these particular laws exist rather than alternatives. What mechanism, if any, selects one set of laws over another?

1.2 The Origin of Physical Laws

The hypothesis that physical laws themselves may emerge from more primitive dynamics is not new. Wheeler’s “it from bit” proposal suggested that information may be more fundamental than matter-energy [13]. Zurek’s quantum Darwinism explores how certain quantum states are selected for

their robustness and information-carrying capacity [14]. Tegmark’s Mathematical Universe Hypothesis posits that mathematical structures exist independently and physical reality corresponds to specific mathematical patterns [11]. Davies has argued for the emergence of laws from deeper principles [4], while Smolin’s cosmological natural selection proposes that universes evolve through black hole production, selecting laws that favor black hole formation [9].

What distinguishes the Theory of the Question of Existence (TQE) from these approaches is its provision of a *quantitative, falsifiable mechanism* for law emergence. Rather than invoking anthropic filtering across infinite universes or appealing to mathematical Platonism, TQE proposes that stable physical laws crystallize from the fundamental coupling of energy and information during cosmogenesis. This coupling creates a probabilistic selection mechanism that preferentially weights law-compatible configurations, leading to the stabilization of complexity-permitting laws.

1.3 TQE Framework Overview

TQE posits that the universe originates from a quantum fluctuation in a *pre-law state*—a regime devoid of classical space-time and established physical constants. In this pre-law phase, the system is governed by a probabilistic modulation rule applied to the space of possible law-states. The modulation depends on two fundamental quantities: vacuum energy fluctuations E and an information-theoretic orientation parameter $I \in [0, 1]$ that measures the degree to which a configuration supports complexity and structure formation.

The core mathematical structure takes the form:

$$P'(\psi) = P(\psi) \cdot f(E, I), \tag{1}$$

where $P(\psi)$ represents the probability distribution over possible law-states ψ , and $f(E, I)$ is a coupling function that biases the selection toward configurations with certain energy-information properties. (The normalized form of this equation is presented in Section 2.1; see equation (2).) The coupling function creates a narrow stability window (also called a “Goldilocks zone”) in parameter space, allowing only universes with the correct balance of energy and information to stabilize their laws through a lock-in mechanism.

This framework reframes the fine-tuning problem: rather than asking why the laws are fine-tuned *a priori*, TQE asks whether fine-tuned laws emerge *dynamically* through a selection process operating during cosmogenesis. The mechanism is probabilistic rather than deterministic, offering a natural explanation for the apparent rarity of complexity-permitting universes.

1.4 TQE as a Conceptual Framework

TQE should be understood as a **conceptual framework** rather than a complete theory. Like early formulations of quantum mechanics (Bohr’s correspondence principle) or general relativity (Einstein’s equivalence principle), TQE provides a *theoretical architecture* within which more detailed theories can be developed.

The framework’s value lies in:

1. Reframing the fine-tuning problem as a dynamical question rather than an anthropic selection,
2. Providing a mathematical structure for law emergence that is both quantitative and falsifiable,
3. Generating testable predictions despite incomplete theoretical definitions,
4. Opening new research directions in quantum cosmology and the origin of physical laws.

I emphasize that TQE is a *framework proposal*, not a final theory. The information parameter I and certain aspects of the coupling function $f(E, I)$ remain incompletely defined, and this manuscript explicitly acknowledges these limitations while demonstrating that the framework’s structure nevertheless enables meaningful theoretical and potentially observational progress.

1.5 Scope and Organization

In this manuscript, I present the theoretical foundations and mathematical structure of TQE, focusing exclusively on the framework itself rather than numerical simulations or detailed comparisons with observational data. I deliberately exclude simulation results and numerical predictions, as these represent a separate phase of research that builds upon the theoretical framework presented here.

The paper is organized as follows. Section 2 develops the theoretical framework, including the core hypothesis, the coupling function, the information parameter (presented as an open question with candidate definitions), the pre-law phase, and the law lock-in mechanism. Section 3 presents the mathematical structure in detail, including probability space definitions, normalization requirements, stability conditions, and conceptual connections to standard physics. Section 4 provides physical interpretation of the framework, discussing energy-information coupling, the cosmogenesis mechanism,

and relations to quantum mechanics and general relativity. Section 5 discusses the framework’s place in the broader landscape of theoretical physics, relations to other approaches, and open questions. Section 6 summarizes the main conclusions.

2 Theoretical Framework

2.1 Core Hypothesis

The Theory of the Question of Existence (TQE) is founded on a fundamental premise: *the origin of the universe does not require an external creator or agency, but arises from an internal mechanism operating within a pre-law quantum state*. This premise implies that the universe originates from a pre-law quantum state—a regime devoid of pre-existing physical laws and classical space-time structure.

In this framework, energy is not merely a passive quantity but carries an intrinsic informational property: a directionality toward complexity. This informational orientation, denoted $I \in [0, 1]$, is postulated as a fundamental property of energy itself, not as an external field or parameter.

2.1.1 Theoretical Status of the Information Parameter

A central component of TQE is the information parameter $I \in [0, 1]$, which measures orientation toward complexity. **I acknowledge that the fundamental definition of I remains an open question.** This is not a deficiency to be hidden, but rather an honest recognition of the framework’s current theoretical limits. The framework’s predictive structure does not require a complete definition of I to make falsifiable predictions about cosmological observations; its value lies in the conceptual architecture and testable consequences, not in providing a final theory.

Several candidate definitions exist, which I discuss in detail in Section 2.3:

- **Holographic entropy-based:** $I \sim S_{\text{entanglement}}/S_{\text{max}}$, where $S_{\text{entanglement}}$ is entanglement entropy and S_{max} is the maximum possible entropy.
- **Structural complexity:** $I = S_{\text{struct}}/S_{\text{max}}$, where S_{struct} measures structural complexity (e.g., pattern formation, hierarchical organization).

- **Computational complexity:** Based on computational complexity theory, measuring the information content required to describe a configuration.
- **Emergent from coupling:** I may not be fundamental but emergent from the energy-information coupling dynamics itself.

The key insight is that *TQE's predictive structure does not require a complete definition of I* to make falsifiable predictions about cosmological observations. The framework's value lies in its conceptual architecture and testable consequences, not in providing a final theory.

2.1.2 The Pre-law Phase

A critical aspect of TQE is the concept of a *pre-law phase*—a regime in which classical space-time, as described by General Relativity (GR), does not exist. In this phase, there is no metric tensor, no geodesic structure, no Einstein field equations, and therefore no space-time curvature in the classical sense. The Hawking-Penrose singularity theorems [6, 7], which prove the inevitability of singularities under GR's assumptions, do not apply in this regime because their prerequisites (classical space-time structure, energy conditions, focusing conditions) are not satisfied.

Relation to Quantum Gravity Proposals: The pre-law phase in TQE shares conceptual similarities with approaches in quantum gravity that treat space-time as emergent. The Wheeler-DeWitt equation of canonical quantum gravity describes a timeless wave function of the universe, operating in a configuration space without classical time. Similarly, loop quantum gravity, causal dynamical triangulation, and string theory approaches propose that space-time emerges from more fundamental structures. TQE's pre-law phase differs in that it operates in a space of *law-states* Ψ rather than geometric configurations, and the selection mechanism (via $f(E, I)$) is explicitly information-theoretic rather than purely geometric or field-theoretic. The pre-law phase can be understood as a regime where neither GR nor standard quantum field theory applies, requiring a new theoretical framework—which TQE provides through the probabilistic modulation rule.

The pre-law phase is instead governed by a probabilistic field $P(\psi)$ defined over the space Ψ of possible law-states ψ . Each $\psi \in \Psi$ represents a candidate configuration of physical laws, constants, and symmetries. The system evolves according to a modulation rule that depends on energy and information:

Definition 2.1 (Core Modulation Rule). The probability distribution $P(\psi)$

over law-states $\psi \in \Psi$ evolves according to:

$$P'(\psi) = \frac{P(\psi) f(E(\psi), I(\psi))}{Z[P]}, \quad (2)$$

where $E(\psi) \in \mathbb{R}$ is the energy associated with state ψ , $I(\psi) \in [0, 1]$ is the information parameter, $f(E, I)$ is the coupling function, and $Z[P]$ is the partition function ensuring normalization:

$$Z[P] = \int_{\Psi} P(\phi) f(E(\phi), I(\phi)) d\phi. \quad (3)$$

2.1.3 Interpretation of Components

The modulation rule (2) encodes several key assumptions:

- **Probability Space Ψ :** The configuration space Ψ represents all admissible microstates or law-states. This space may include candidate sets of physical constants, symmetry groups, conservation laws, and interaction structures. The precise structure of Ψ depends on the specific implementation, but it must be sufficiently rich to encode the diversity of possible physical laws.
- **Energy Functional $E : \Psi \rightarrow \mathbb{R}$:** Each state ψ is associated with an energy value $E(\psi)$, which I interpret as vacuum fluctuation energy. In the pre-law phase, I model these energy values by a heavy-tailed distribution (e.g., log-normal), allowing for rare large fluctuations that could initiate cosmogenesis.
- **Information Parameter $I : \Psi \rightarrow [0, 1]$:** The information parameter $I(\psi)$ quantifies how well-aligned state ψ is with an information-bearing direction toward complexity. The definition of I is a central open question in TQE (see Section 2.3). Conceptually, states with higher I values are those that better support structure formation, complexity, and the emergence of coherent dynamics.
- **Coupling Function $f(E, I)$:** This function implements the selection bias, preferentially weighting states with certain energy-information combinations. The explicit form is discussed in Section 2.2.

2.1.4 Non-linearity and Functional Dependence

The partition function $Z[P]$ is a *functional* of the probability distribution P , not merely a function of parameters. When $I(\psi)$ depends on the evolving distribution P (e.g., through information-theoretic measures computed between

consecutive probability distributions), the functional dependence creates an implicit equation for P' :

$$P'(\psi) = \frac{P(\psi) f(E(\psi), I(\psi; P'))}{Z[P']}, \quad (4)$$

where $Z[P'] = \int_{\Psi} P(\phi) f(E(\phi), I(\phi; P')) d\phi$. This equation must be solved self-consistently (e.g., via fixed-point iteration: $P^{(n+1)} = \mathcal{E}[P^{(n)}]$). The existence and uniqueness of solutions is an open question requiring further analysis.

When $I(\psi)$ is independent of P (e.g., in the inherent information model), the equation becomes explicit and $Z[P]$ is simply a normalization constant. However, when I depends on P (e.g., via KL divergence between P_t and P_{t+1}), the map $P \mapsto P'$ becomes a non-linear operator on probability measures, creating a fixed-point problem.

This non-linearity is crucial: it allows for feedback loops where the current distribution influences the information parameter, which in turn modulates the distribution evolution. The resulting dynamics can exhibit rich behavior, including phase transitions, bifurcations, and attractors corresponding to stable law configurations.

To ensure $P'(\psi) \geq 0$ for all ψ , I require either that $1 + \alpha I(\psi) \geq 0$ for all ψ (when using a linear information term) or that the information channel be exponentiated as $\exp[\beta I(\psi)]$ to guarantee positivity. These choices represent different modeling assumptions and can be tested through their implications for law selection.

2.2 The Coupling Function

The coupling function $f(E, I)$ is the mathematical embodiment of TQE's central hypothesis: that stable physical laws emerge from the joint influence of energy and information. This function implements a selection bias that preferentially weights certain configurations in the space of possible law-states.

2.2.1 Explicit Form

I adopt the following functional form for the coupling function:

$$f(E, I) = \exp \left[-\frac{(E - E_c)^2}{2\sigma^2} \right] (1 + \alpha I), \quad (5)$$

where:

- $E \in \mathbb{R}$ is the vacuum fluctuation energy associated with a state,
- $I \in [0, 1]$ is the information parameter (discussed in Section 2.3),
- $E_c \in \mathbb{R}$ is the *Goldilocks energy*—the center of the stability window,
- $\sigma > 0$ is the *stability width*—the tolerance around E_c ,
- $\alpha \geq 0$ is the *orientation bias strength*—quantifies the effect of information.

2.2.2 Physical Interpretation of Components

The coupling function (5) consists of two multiplicative factors, each encoding a distinct physical hypothesis:

1. Energetic Stability Window (Gaussian Term)

The exponential factor $\exp[-(E - E_c)^2/(2\sigma^2)]$ creates a narrow stability window (a “Goldilocks zone”) centered at E_c with width controlled by σ . This term encodes the hypothesis that stable physical laws can emerge only when the vacuum fluctuation energy falls within a specific range—not too low (insufficient to initiate cosmogenesis) and not too high (unstable, chaotic configurations).

Physically, E_c represents the critical energy scale around which universes can stabilize their laws. The parameter σ controls how sensitive the system is to deviations from this optimal energy. A larger σ broadens the stability window, making stabilization more probable but potentially less precise. A smaller σ makes the window narrower, requiring more precise energy values but potentially yielding more stable configurations.

The Gaussian form is a convenient choice that provides smooth weighting and mathematical tractability. Alternative functional forms (e.g., top-hat functions, Lorentzian profiles) could be explored, though the qualitative behavior would remain similar: preferential selection of states near E_c .

2. Information Bias (Linear Term)

The linear factor $(1 + \alpha I)$ implements the hypothesis that states with higher information content are preferentially selected. When $I = 0$, this factor equals unity, and the selection depends purely on the energetic term. When $I > 0$, states receive enhanced weight proportional to αI , biasing the selection toward configurations that support complexity.

The parameter α quantifies the strength of this informational bias. For $\alpha = 0$, information plays no role, and the system reduces to pure energetic selection. For $\alpha > 0$, information becomes increasingly important in determining which law-states are selected. The linear form $(1 + \alpha I)$ ensures that

when $I = 0$, the information term contributes neutrally, while when $I \rightarrow 1$, the maximum enhancement factor is $(1 + \alpha)$.

2.2.3 Candidate Coupling Functions

While the exact form of $f(E, I)$ depends on the definition of I , I can explore several candidate structures. Each formulation generates different selection windows, but all share the property of creating a viable parameter range in (E, I) space.

1. Gaussian-Exponential Coupling (Primary Formulation)

This is the primary formulation, defined earlier in equation (5):

$$f(E, I) = \exp \left[-\frac{(E - E_c)^2}{2\sigma^2} \right] (1 + \alpha I), \quad (6)$$

where $\alpha \geq 0$ controls the information bias strength. This formulation is interpretable and allows independent control of energetic and informational selection.

2. Exponential Coupling

For stronger informational biases, one may replace the linear term with an exponential form:

$$f(E, I) = \exp \left[-\frac{(E - E_c)^2}{2\sigma^2} + \beta I \right], \quad (7)$$

where $\beta \geq 0$ controls the information bias strength. This formulation guarantees $f(E, I) > 0$ for all E, I and allows for stronger biases, but it modifies the interaction between the energetic and informational terms from multiplicative to additive in the exponent.

3. Power-Law Coupling

An alternative formulation uses a power-law dependence on information:

$$f(E, I) = \exp \left[-\frac{(E - E_c)^2}{2\sigma^2} \right] \cdot I^\alpha, \quad (8)$$

where $\alpha > 0$ is an exponent. This form ensures that states with $I = 0$ receive zero weight, making information orientation strictly necessary for selection.

4. Threshold Coupling

A threshold-based formulation uses a step function:

$$f(E, I) = \Theta(I - I_{\text{crit}}) \cdot \exp \left[-\frac{(E - E_c)^2}{2\sigma^2} \right] \cdot g(E), \quad (9)$$

where Θ is the Heaviside step function, I_{crit} is a critical information threshold, and $g(E)$ is an energy-dependent function. This form encodes a sharp transition: only states with $I > I_{\text{crit}}$ can be selected.

All these candidate formulations are compatible with TQE’s core hypothesis. The choice depends on the desired behavior and the specific definition of I adopted. The primary formulation (5) is used throughout this manuscript for its interpretability and mathematical tractability.

2.2.4 Normalization and Probability Conservation

The normalized update rule (2) ensures that probability is conserved:

$$\int_{\Psi} P'(\psi) d\psi = \frac{\int_{\Psi} P(\psi) f(E(\psi), I(\psi)) d\psi}{Z[P]} = 1, \quad (10)$$

since $Z[P] = \int_{\Psi} P(\phi) f(E(\phi), I(\phi)) d\phi$ by definition.

The partition function $Z[P]$ plays a crucial role: it normalizes the biased distribution, ensuring that the selection mechanism does not violate probability conservation. Formally, $Z[P]$ is a *functional* of P because $I(\psi)$ may itself depend on the distribution P . This functional dependence makes the update map $P \mapsto P'$ a non-linear operator on the space of probability measures.

2.2.5 Parameter Interpretation

The parameters (E_c, σ, α) (or (E_c, σ, β) in the exponential form) are not arbitrary fitting constants but structural elements of the model encoding physical assumptions:

- **E_c (Goldilocks Energy):** This parameter defines the energy scale at which stable law formation is most probable. It may be related to fundamental scales in physics (e.g., Planck scale, vacuum energy scale), though the precise relationship remains to be determined.
- **σ (Stability Width):** This parameter quantifies how broad the stability window is. A larger σ implies that a wider range of energies can lead to stable universes, potentially making cosmogenesis more common. A smaller σ implies stricter energy requirements, making stable universes rarer but potentially more robust.
- **α or β (Orientation Bias Strength):** This parameter quantifies the importance of information in law selection. When small, energy dominates; when large, information plays a crucial role in determining which configurations are selected.

The precise values of these parameters and their relationship to measurable physical quantities remain open questions that require further theoretical development and, potentially, comparison with cosmological observations.

2.3 The Information Parameter

The information parameter $I \in [0, 1]$ plays a central role in TQE, quantifying the degree to which a configuration supports complexity and structure formation. However, *the fundamental definition of I remains an open question* that requires further theoretical development. This section presents several operational definitions that have been explored, while acknowledging that the ultimate theoretical foundation remains to be established.

Open Question 2.1 (Fundamental Definition of Information Parameter). What is the fundamental definition of the information parameter I ? How can it be derived from first principles in quantum field theory, information theory, or holographic principles? Which operational definition best captures the “orientation toward complexity” that TQE postulates?

2.3.1 Operational Definitions

While the fundamental definition of I is not yet established, several operational definitions have been explored. These should be understood as *candidate formulations* rather than definitive answers:

1. Structural Complexity Based (Heuristically Proposed)

I propose a heuristically motivated definition based on structural complexity:

$$I(\psi) = \frac{S_{\text{struct}}(\psi)}{S_{\text{max}}}, \quad (11)$$

where $S_{\text{struct}}(\psi)$ measures the structural complexity of state ψ (e.g., pattern formation, hierarchical organization, coherent dynamics), and S_{max} is the maximum possible structural complexity. This definition ensures $0 \leq I \leq 1$ and directly captures the “orientation toward complexity” that TQE postulates. The structural complexity S_{struct} could be quantified through measures such as:

- Pattern formation metrics (e.g., correlation length, order parameters),
- Hierarchical organization measures (e.g., nested structure, scale separation),
- Coherent dynamics indicators (e.g., phase coherence, synchronization),

- Information-theoretic complexity (e.g., effective complexity, algorithmic information content).

While this definition is heuristically motivated rather than derived from first principles, it provides a concrete operational framework for computing I in specific contexts.

2. Kullback-Leibler Divergence Based

An alternative operational definition computes I from consecutive probability distributions P_t and P_{t+1} via normalized Kullback-Leibler (KL) divergence:

$$I = \frac{D_{\text{KL}}(P_t \parallel P_{t+1})}{1 + D_{\text{KL}}(P_t \parallel P_{t+1})}, \quad (12)$$

where

$$D_{\text{KL}}(P_t \parallel P_{t+1}) = \int_{\Psi} P_t(\psi) \ln \left[\frac{P_t(\psi)}{P_{t+1}(\psi)} \right] d\psi. \quad (13)$$

This definition ensures $0 \leq I \leq 1$ and measures the information-theoretic distance between consecutive distributions. Intuitively, larger KL divergence corresponds to greater change in the distribution, which may signal orientation toward complexity. However, this is an operational definition that depends on the temporal evolution of P itself, creating a feedback loop that requires careful analysis.

2. Shannon Entropy Combination

An alternative approach combines KL divergence with Shannon entropy:

$$I = I_{\text{KL}} \times I_{\text{Shannon}}, \quad (14)$$

where I_{KL} is computed via (12) and I_{Shannon} is derived from the Shannon entropy:

$$S[P] = - \int_{\Psi} P(\psi) \ln P(\psi) d\psi. \quad (15)$$

This combination attempts to capture both the asymmetry between distributions (via KL divergence) and the intrinsic complexity (via entropy). However, the precise functional form of this combination remains arbitrary.

3. Emergent Information Model

Another candidate formulation treats I as emerging dynamically from energy fluctuations:

$$I_{t+1} = \gamma I_t + \alpha |\Delta E_t| + \beta \text{corr}(\Delta E_t, \Delta E_{t-1}), \quad (16)$$

where $\Delta E_t = E_t - E_{t-1}$ is the energy change, $\gamma \in [0, 1]$ is a decay factor, $\alpha, \beta \geq 0$ are weights, and corr denotes correlation.

This formulation encodes a self-referential learning mechanism where I depends on its own past value (memory term γI_t), immediate energy fluctuations ($\alpha|\Delta E_t|$), and pattern detection in energy changes ($\beta \text{ corr}$). While physically intuitive, this model introduces additional parameters and requires normalization to ensure $I \in [0, 1]$.

4. Inherent Information Model

An alternative view treats I as an inherent property of energy states:

$$I = C \cdot f(E/E_0), \quad (17)$$

where $C > 0$ is a normalization constant, $E_0 > 0$ is a reference energy scale, and f is a function such as $\ln(E/E_0)$ or $(E/E_0)^\gamma$ for some exponent γ .

This model posits that information is directly encoded in the energy scale, growing with energy (logarithmically or power-law). While simple, this formulation does not capture the temporal evolution or self-referential aspects that may be crucial for law emergence.

5. Threshold Information Model

A third candidate formulation uses a threshold activation:

$$I = \begin{cases} 0 & \text{if } E < E_c, \\ I_0 + \lambda(E - E_c) & \text{if } E \geq E_c, \end{cases} \quad (18)$$

where E_c is a critical energy threshold, $I_0 \geq 0$ is a baseline value, and $\lambda > 0$ is a slope parameter.

This model encodes a sharp transition at critical energy, suggesting that information “activates” only above a certain threshold. While providing a clear physical interpretation, the threshold model lacks the smoothness and self-referential behavior that may be necessary for gradual law emergence.

2.3.2 Open Questions and Future Directions

The existence of multiple candidate definitions highlights that the fundamental nature of the information parameter I remains unresolved. Several open questions require further investigation:

Open Question 2.2 (Information Parameter Foundation). Can the information parameter I be derived from fundamental principles in quantum field theory, quantum information theory, or holographic approaches? Is it related to entanglement entropy, quantum Fisher information, or other information-theoretic quantities?

Open Question 2.3 (Optimal Operational Definition). Which operational definition best captures the “orientation toward complexity” that TQE postulates? Can multiple definitions be unified under a single theoretical framework?

Open Question 2.4 (Dependence on Distribution). Should I depend on the evolving probability distribution P (creating feedback loops) or be defined independently of P ? What are the implications of each choice for law emergence?

2.3.3 Physical Interpretation

Despite the open questions regarding its precise definition, the information parameter I is conceptually clear: it quantifies how well-aligned a configuration is with an information-bearing direction toward complexity. States with higher I values are those that better support:

- Structure formation and self-organization,
- Coherent dynamics and pattern emergence,
- Complexity growth and hierarchical organization,
- Stable, reproducible behavior.

In the context of law selection, I acts as a “guide” that preferentially weights configurations that support these properties, steering the system toward complexity-permitting law sets.

2.3.4 Properties and Constraints

Regardless of the specific definition, the information parameter must satisfy certain constraints:

1. **Range:** $I \in [0, 1]$ to ensure boundedness and interpretability.
2. **Normalization:** The definition must allow computation of $I(\psi)$ for any state $\psi \in \Psi$.
3. **Positivity:** In the coupling function (5), I require $1 + \alpha I \geq 0$, which is automatically satisfied for $\alpha \geq 0$ and $I \in [0, 1]$.
4. **Continuity (desirable):** Smooth dependence on state parameters may be necessary for stable evolution, though threshold models may intentionally violate this.

The search for the fundamental definition of I represents one of the most important open questions in TQE, as it directly affects the framework’s predictive power and theoretical completeness.

2.3.5 Possible Observational Signatures of I

Even without a complete definition, I may leave observable signatures in cosmological data. These signatures are *model-dependent* but suggest paths toward making I empirically constrained rather than purely theoretical:

- **CMB anomalies:** Non-Gaussianities in the cosmic microwave background (CMB) reflecting I 's structure, such as anomalous alignments in multipoles (e.g., “Axis of Evil”) or hemispherical power asymmetries.
- **Large-scale structure:** Patterns in galaxy distribution and large-scale structure formation that deviate from random Gaussian fluctuations, potentially encoding information about the selection mechanism.
- **Primordial gravitational waves:** Constraints on the tensor-to-scalar ratio and primordial gravitational wave spectrum that may reflect the energy-information coupling during cosmogenesis.
- **Statistical patterns:** Non-random statistical features in cosmological observables that distinguish our universe from random realizations, consistent with the stabilization dynamics.

These potential signatures provide a pathway for empirical validation of TQE, even while the fundamental definition of I remains open. The framework's testability does not require a complete theoretical definition of I ; rather, observational constraints can inform and guide the theoretical development of I 's definition.

2.4 Pre-law Phase

A fundamental aspect of TQE is the concept of a *pre-law phase*—a regime that precedes the emergence of classical space-time and established physical laws. This phase is characterized by the absence of the structures and constraints that define standard General Relativity (GR).

2.4.1 Absence of Classical Space-Time

In the pre-law phase, classical space-time, as described by GR, does not exist. Specifically:

- **No metric tensor:** There is no $g_{\mu\nu}(x)$ that defines space-time distances and geometry.

- **No geodesic structure:** The concept of geodesics (shortest paths in space-time) is not defined.
- **No Einstein field equations:** The relation $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ cannot be formulated.
- **No space-time curvature:** In the absence of a metric, space-time curvature has no meaning.

This absence is crucial: it implies that the pre-law phase lies *outside the domain of applicability of GR*, and therefore GR’s theorems and predictions do not necessarily apply.

2.4.2 Singularity Theorems Do Not Apply

The Hawking-Penrose singularity theorems [6, 7] prove the inevitability of singularities under GR’s assumptions. However, these theorems require:

1. A classical, well-defined space-time manifold with metric $g_{\mu\nu}$,
2. Energy conditions on the stress-energy tensor $T_{\mu\nu}$,
3. Focusing conditions on geodesic congruences,
4. Global hyperbolicity and smooth metric structure.

In the pre-law phase, *none of these prerequisites are satisfied*. Therefore, the singularity theorems’ assumptions fail, and their conclusions (the inevitability of singularities) do not necessarily hold.

Remark 2.1. The strength of the singularity theorems derives precisely from their assumptions. If these assumptions fail in the early universe—as TQE postulates for the pre-law phase—then singularities are not mathematically necessary. The “initial singularity” should be understood not as a physical entity but as the limit of GR’s extrapolation backward in time.

This observation reframes the problem: rather than accepting an initial singularity as an inescapable consequence of GR, TQE proposes that GR itself emerges from a pre-law regime where singularity concepts do not apply.

2.4.3 Probabilistic Field Governing

In the absence of classical space-time, the pre-law phase is governed by a *probabilistic field* $P(\psi)$ defined over the space Ψ of possible law-states. This field evolves according to the modulation rule (2), which depends on energy

and information but not on space-time coordinates (which do not exist in this phase).

The system is characterized by:

- **Quantum fluctuations:** Vacuum energy fluctuations can occur with amplitudes determined by quantum uncertainty principles. Large fluctuations, enabled by the uncertainty relation $\Delta E \cdot \Delta t \gtrsim \hbar/2$, can initiate cosmogenesis.
- **Probabilistic selection:** The modulation rule (2) preferentially weights certain configurations, steering the system toward law-compatible states.
- **No pre-existing laws:** Physical constants, symmetries, and conservation laws are not fixed but are subject to selection through the probabilistic mechanism.

2.4.4 Transition to Classical Regime

The transition from the pre-law phase to a law-governed classical regime occurs through the *lock-in mechanism* (discussed in Section 2.5). Once law lock-in occurs, classical space-time emerges, and GR becomes applicable. This emergence is not a fundamental feature of the universe but rather a consequence of the selection process operating in the pre-law phase.

The key insight is that *space-time itself is emergent*—it arises from the selection of law-compatible configurations, not as a pre-existing structure. This perspective aligns with approaches in quantum gravity that treat space-time as emergent (e.g., loop quantum gravity, causal dynamical triangulation, string theory), though TQE provides a different mechanism for space-time emergence through law selection.

2.4.5 Uncertainty and Large Fluctuations

While detailed discussion of quantum uncertainty is beyond the scope of this work, I note that in the pre-law phase, the uncertainty principle allows for large energy excursions on short timescales. These fluctuations are not constrained by established physical laws (which do not yet exist) and can therefore reach magnitudes sufficient to initiate cosmogenesis.

The uncertainty relation $\Delta E \cdot \Delta t \gtrsim \hbar/2$ should be understood not merely as a measurement limit but as a structural trait of the pre-law regime. Here I use this heuristic time–energy uncertainty relation as a structural lower bound on fluctuation windows, not as an operator-based relation (since time

is a parameter, not an operator, in standard quantum mechanics). It guarantees a lower bound on fluctuations: perfect rest is impossible, and large deviations from equilibrium can occur.

When such a fluctuation couples with information orientation (through the coupling function $f(E, I)$), it can trigger the selection of law-compatible configurations, initiating the transition to a law-governed universe. The precise threshold at which this transition occurs depends on the parameters (E_c, σ, α) and the definition of I , which remain open questions.

2.5 Law Lock-in Mechanism

The transition from the pre-law phase to a law-governed classical regime occurs through a *lock-in mechanism*—a process by which a quasi-stationary set of physical laws stabilizes and becomes effectively permanent. I will refer to this stabilization process as “law lock-in” throughout this manuscript. This mechanism is the central dynamical feature that transforms the probabilistic superposition of law-states into a concrete, stable configuration.

2.5.1 The Lock-in Criterion

Law lock-in is operationally defined as the stabilization of key parameters to within a specified tolerance. Specifically, a universe is said to have achieved law lock-in when the relative change in key observables satisfies:

$$\frac{\Delta O}{O} < \epsilon \quad (19)$$

over at least n consecutive epochs, where O denotes a key observable (e.g., fundamental constants, coupling strengths), $\epsilon > 0$ is a small threshold (e.g., $\epsilon = 5 \times 10^{-3}$), and $n \geq 6$ is the minimum number of consecutive stable epochs.

This criterion distinguishes universes that stabilize their laws from those that remain in a chaotic or transitional state. The choice of ϵ and n determines the strictness of the lock-in condition: smaller ϵ or larger n implies stricter requirements for stability.

2.5.2 Complexity Parameter and Stability Window

The lock-in mechanism depends critically on the *complexity parameter*:

$$X = E \cdot I, \quad (20)$$

which represents the product of energy and information. This parameter functions as a *stability gate*: universes must have X values within a narrow viable parameter range to have any chance of stabilizing their laws.

This stability window is not a fixed range but emerges dynamically from the interaction of the coupling function (5) with the probability distribution $P(\psi)$. The Gaussian factor $\exp[-(E - E_c)^2/(2\sigma^2)]$ creates a preferred energy range around E_c , while the information term $(1 + \alpha I)$ enhances states with higher I values. The product $X = E \cdot I$ must balance these two factors: too small X (low energy or low information) cannot initiate stabilization, while too large X (extreme energy or information) leads to instability.

Remark 2.2. The stability window for X is analogous to critical points in condensed matter physics (e.g., superconducting transition temperature T_c). It marks the boundary between chaotic and stable regimes, emerging from the collective dynamics rather than being imposed a priori.

2.5.3 Asymmetry and Lock-in Trigger

While the complexity parameter $X = E \cdot I$ governs the *possibility* of stability, the *achievement* of lock-in depends on the asymmetry parameter:

$$|E - I|. \quad (21)$$

Universes with large asymmetry $|E - I| \gg 0$ are more likely to achieve rapid and decisive law lock-in, while universes with nearly balanced $E \approx I$ may stabilize but are less likely to fully lock in their laws. This asymmetry acts as a *lock-in trigger*, breaking symmetry and propelling the system toward a single, unchanging configuration.

Remark 2.3. This two-factor mechanism (stability gate via $X = E \cdot I$, lock-in trigger via $|E - I|$) explains why the most successful universes may not lie at the peak of the stability probability distribution. Rather, they represent an optimal compromise: stable enough to survive the chaotic phase, yet asymmetric enough to trigger rapid finalization of laws.

2.5.4 Process of Lock-in

The lock-in process can be understood as follows:

1. Exploration Phase: The system explores the space Ψ of possible law-states under the influence of the modulation rule (2). Energy fluctuations provide diversity, while the coupling function biases selection toward certain configurations.

2. Convergence Phase: As the system evolves, configurations with favorable (E, I) combinations gain statistical weight. The probability distribution $P(\psi)$ begins to concentrate around law-compatible states.

3. Threshold Crossing: When $X = E \cdot I$ enters the viable parameter range and $|E - I|$ exceeds a threshold, the system crosses a critical point. The probability distribution rapidly concentrates on a narrow subset of configurations.

4. Stabilization: The selected configuration exhibits persistent stability, with key parameters changing by less than ϵ over n consecutive epochs. At this point, law lock-in is achieved.

5. Classical Expansion: Once laws are locked in, the system transitions to classical expansion. Space-time emerges, GR becomes applicable, and the universe evolves according to the stabilized laws.

2.5.5 Emergence of Classical Space-Time

An important consequence of law lock-in is the *emergence of classical space-time*. Prior to lock-in, there is no metric, no geodesics, and no space-time structure. After lock-in, the selected law configuration includes a metric structure (encoded in the law-state ψ), allowing classical space-time to emerge.

This emergence is not imposed externally but arises naturally from the selection process. The metric $g_{\mu\nu}$ is part of the law-state ψ that was selected through the probabilistic mechanism. Once locked in, this metric structure becomes permanent (or at least extremely stable), and GR governs the subsequent evolution.

Remark 2.4. The emergence of space-time from law selection provides a mechanism for connecting the pre-law TQE phase to the classical GR universe. However, the precise mathematical details of how the metric structure is encoded in ψ and how it emerges remain open questions requiring further development.

2.5.6 Irreversibility and Stability

Once law lock-in occurs, the stabilized laws become effectively permanent. The system has crossed a phase transition from a probabilistic superposition to a classical, deterministic evolution. This irreversibility explains why physical laws appear immutable: they were selected through a probabilistic mechanism but, once locked in, become the foundation for all subsequent dynamics.

The stability of locked-in laws is not absolute but rather statistical: deviations from the selected configuration are exponentially suppressed by the coupling function $f(E, I)$. Large deviations that could destabilize the laws are extremely unlikely once lock-in is achieved, explaining why new universes do not form within an existing universe (the laws suppress large fluctuations).

This mechanism provides a natural explanation for why physical laws appear constant throughout cosmic history: they were selected early in cosmogenesis and have remained stable since then due to the lock-in mechanism.

3 Mathematical Structure

This section develops the mathematical foundations of TQE in detail, focusing on the probabilistic structure, normalization requirements, stability conditions, and conceptual connections to standard physics.

3.1 Probability Space and State Evolution

3.1.1 Configuration Space

The configuration space Ψ represents the set of all admissible microstates or law-states. Each element $\psi \in \Psi$ encodes a candidate configuration of physical laws, including:

- Physical constants (e.g., fundamental coupling constants, masses, cosmological constant),
- Symmetry groups (e.g., gauge groups, Lorentz group),
- Conservation laws (e.g., energy, momentum, charge conservation),
- Interaction structures (e.g., gauge interactions, gravitational coupling),
- Space-time structure (e.g., metric, dimensionality, topology).

The precise mathematical structure of Ψ depends on the specific implementation. It may be a finite or infinite set, a continuous manifold, or a more general space. For definiteness, I assume Ψ is equipped with a measure structure that allows integration over law-states.

3.1.2 Energy and Information Functionals

Each state $\psi \in \Psi$ is associated with two functionals:

Energy Functional: $E : \Psi \rightarrow \mathbb{R}$ assigns an energy value $E(\psi)$ to each state. This energy represents vacuum fluctuation energy. I model the pre-law vacuum energy fluctuations by a heavy-tailed distribution (e.g., log-normal) to allow rare large excursions. Large values of $E(\psi)$ correspond to rare, high-energy fluctuations that could initiate cosmogenesis.

Information Functional: $I : \Psi \rightarrow [0, 1]$ assigns an information value $I(\psi) \in [0, 1]$ to each state. As discussed in Section 2.3, the definition of I remains an open question. Critically, I may depend on the evolving probability distribution P itself, creating a feedback loop where $I(\psi) = I(\psi; P)$.

3.1.3 Probability Measure Evolution

The probability distribution $P(\psi)$ over Ψ evolves according to the modulation rule (2). The update map $P \mapsto P'$ is a *non-linear operator* on the space of probability measures because:

1. The partition function $Z[P]$ is a functional of P ,
2. The information parameter $I(\psi; P)$ may depend on P ,
3. The coupling function $f(E(\psi), I(\psi; P))$ therefore depends on P .

This non-linearity is crucial: it allows for rich dynamical behavior, including phase transitions, bifurcations, and attractors corresponding to stable law configurations.

3.1.4 Evolution Operator

The update rule (2) defines an evolution operator $\mathcal{E}$ on the space of probability measures:

$$P'(\psi) = \mathcal{E}[P](\psi) = \frac{P(\psi) f(E(\psi), I(\psi; P))}{Z[P]}. \quad (22)$$

The operator $\mathcal{E}$ is non-linear and depends on the current distribution P through both $Z[P]$ and $I(\psi; P)$. This makes the evolution equation a non-linear functional equation.

3.1.5 Fixed Points and Stability

A *fixed point* of the evolution operator $\mathcal{E}$ is a probability distribution P^* that satisfies:

$$P^*(\psi) = \mathcal{E}[P^*](\psi) = \frac{P^*(\psi) f(E(\psi), I(\psi; P^*))}{Z[P^*]}. \quad (23)$$

At a fixed point, the probability distribution does not change under the evolution rule. These fixed points correspond to stable law configurations, where the system has reached a quasi-stationary state.

The stability of a fixed point P^* depends on the properties of the evolution operator $\mathcal{E}$ near P^* . Linear stability analysis involves computing the Fréchet derivative of $\mathcal{E}$ at P^* and studying its spectrum. Fixed points with eigenvalues all having negative real parts are stable; those with any eigenvalue having positive real part are unstable.

3.2 Normalization and Partition Function

3.2.1 Partition Function as a Functional

The partition function $Z[P]$ defined in (3) is a *functional* of the probability distribution P , not merely a function of parameters. This functional dependence arises because:

- The integration is over Ψ weighted by P ,
- The information parameter $I(\psi; P)$ may depend on P ,
- Therefore $f(E(\psi), I(\psi; P))$ depends on P .

This makes $Z[P]$ a highly non-trivial object that must be computed self-consistently with the evolution of P .

3.2.2 Properties of the Partition Function

The partition function $Z[P]$ satisfies several important properties:

1. Positivity: Since $P(\psi) \geq 0$ and $f(E, I) > 0$ for all ψ (with appropriate choice of coupling function), it follows that $Z[P] > 0$.

2. Normalization: The partition function ensures that $P'(\psi)$ remains normalized:

$$\int_{\Psi} P'(\psi) d\psi = \frac{\int_{\Psi} P(\psi) f(E(\psi), I(\psi; P)) d\psi}{Z[P]} = \frac{Z[P]}{Z[P]} = 1. \quad (24)$$

3. Functional Dependence: As a functional of P , $Z[P]$ can change dramatically as P evolves. This evolution is coupled to the evolution of P itself, creating feedback loops.

At a fixed point P^* , the normalization condition $\int_{\Psi} P^*(\psi) d\psi = 1$ must hold. This imposes a constraint on the possible fixed points: only distributions that satisfy both the fixed-point equation (23) and normalization can be fixed points.

3.2.3 Renormalization

The evolution rule (2) can be understood as a *renormalization* procedure:

1. Multiply $P(\psi)$ by the coupling function $f(E(\psi), I(\psi; P))$ (biasing),
2. Normalize by dividing by $Z[P]$ (renormalization),
3. Result is a new probability distribution $P'(\psi)$.

This renormalization ensures probability conservation while implementing the selection bias. The procedure is analogous to renormalization in quantum field theory, where bare parameters are rescaled to maintain consistency.

3.3 Stability Conditions

3.3.1 Stability Window for Energy

The coupling function (5) creates a stability window for energy centered at E_c with width controlled by σ . States with $|E - E_c| \ll \sigma$ receive high weight from the Gaussian factor, while states with $|E - E_c| \gg \sigma$ are exponentially suppressed.

The stability window is not a strict constraint but rather a statistical preference. States outside the window can still exist but are exponentially less likely. The width σ controls how strict this preference is: larger σ broadens the window, allowing a wider range of energies to contribute significantly.

3.3.2 Complexity Parameter Threshold

The complexity parameter $X = E \cdot I$ must fall within a viable range for stability. While the precise bounds depend on the specific values of (E_c, σ, α) , qualitatively:

- Too small X : Insufficient energy or information cannot initiate stabilization.

- Optimal X : Balance between energy and information allows stable law emergence.
- Too large X : Extreme values lead to instability and chaos.

The viable parameter range for X emerges dynamically from the interaction of the coupling function with the probability distribution. It is not imposed a priori but arises from the collective dynamics.

3.3.3 Lock-in Conditions

Law lock-in occurs when the system satisfies:

1. **Energy condition:** $|E - E_c| < \sigma$ (within stability window),
2. **Complexity condition:** $X = E \cdot I$ is in viable range,
3. **Asymmetry condition:** $|E - I| > \delta$ for some threshold δ ,
4. **Stability condition:** $\Delta O/O < \epsilon$ over n consecutive epochs, where O is a key observable.

These conditions together determine whether a universe can stabilize its laws. The precise values of the thresholds $(\sigma, \delta, \epsilon, n)$ depend on the specific implementation and remain open questions.

3.3.4 Critical Points as Emergent Properties

The thresholds (E_c, σ, α) are not arbitrary parameters but emerge from the dynamics. They are *critical points* analogous to phase transitions in condensed matter physics (e.g., superconducting transition temperature T_c). These critical points mark the boundary between chaotic and stable regimes.

The emergence of critical points from collective dynamics is a hallmark of complex systems. In TQE, these critical points define the conditions under which stable physical laws can emerge, providing a natural explanation for fine-tuning: only universes that happen to satisfy these conditions can stabilize their laws.

3.4 Connection to Standard Physics

This section discusses the conceptual connection between TQE's master coupling $f(E, I)$ and standard physics. I emphasize that these are *conceptual mappings* rather than rigorous derivations. The precise mathematical derivation of standard physics from TQE remains an open question.

3.4.1 Hierarchical View

TQE proposes a hierarchical view of physics, where the master coupling $f(E, I)$ at “Tier 0” gives rise to standard physics at higher tiers through coarse-graining:

- **Tier 0:** Master coupling $P' = Pf/Z$ (TQE core),
- **Tier 1:** Conservation laws ($f = 1$ for isolated systems),
- **Tier 2:** Thermodynamics (f proportional to Boltzmann weights),
- **Tier 3:** Gravitation (f reweights sources),
- **Tier 4:** Relativity (f rescales effective energy),
- **Tier 5:** Electromagnetism (f biases field configurations),
- **Tier 6:** Quantum mechanics (f appears in Hamiltonian),
- **Tier 7:** Fundamental forces (f captures interaction structure),
- **Tier 8:** Cosmology (f rescales cosmic sources).

At each tier, the effective bias $f_{\text{eff}}^{(k)}$ is obtained by coarse-graining the microscopic coupling $f(E, I)$ over the relevant macro-manifold.

3.4.2 Conceptual Mapping

The connection between TQE and standard physics is *conceptual* rather than mathematically rigorous. I propose that:

- **Conservation laws** arise when $f = 1$ (no selection bias),
- **Thermodynamics** emerges when f is proportional to Boltzmann weights,
- **Gravity** appears when f reweights stress-energy sources,
- **Quantum mechanics** results when f enters the Hamiltonian,
- **Cosmology** follows when f modifies Friedmann equation sources.

These mappings are heuristics that suggest how TQE might connect to standard physics, but they require rigorous derivation to be fully established.

3.4.3 Open Questions

Open Question 3.1 (Derivation of Standard Physics). Can the laws of standard physics (conservation laws, thermodynamics, gravity, quantum mechanics) be rigorously derived from TQE’s master coupling $f(E, I)$ through appropriate coarse-graining procedures? What are the precise mathematical steps required?

Open Question 3.2 (Hierarchical Mapping). How does the microscopic coupling $f(E, I)$ map to effective biases $f_{\text{eff}}^{(k)}$ at different tiers? What is the precise coarse-graining procedure that connects Tier 0 to higher tiers?

The development of a rigorous hierarchical mapping represents one of the most important open questions in TQE, as it would establish a direct connection between the framework and established physics.

4 Physical Interpretation

This section provides physical interpretation of the mathematical structure developed in Sections 2 and 3, connecting the abstract formalism to physical concepts.

4.1 Energy-Information Coupling

The central hypothesis of TQE is that stable physical laws emerge from the coupling of energy and information. This coupling is implemented through the modulation rule (2), where the coupling function $f(E, I)$ creates a selection bias that preferentially weights certain configurations.

4.1.1 The Role of Energy

Energy E in TQE represents vacuum fluctuation energy—the quantum fluctuations in the vacuum that are permitted by uncertainty principles. In the pre-law phase, these fluctuations are not constrained by established physical laws and can reach large magnitudes. The distribution of energy values is heavy-tailed (e.g., log-normal), allowing for rare but significant fluctuations that could initiate cosmogenesis.

The Gaussian factor $\exp[-(E - E_c)^2/(2\sigma^2)]$ in the coupling function creates a stability window for energy, preferentially selecting states with energies near the critical value E_c . This window ensures that only fluctuations with the correct energy scale can initiate stable law formation: too low energy cannot overcome the threshold, while too high energy leads to instability.

4.1.2 The Role of Information

Information I in TQE quantifies the degree to which a configuration supports complexity and structure formation. As discussed in Section 2.3, the fundamental definition of I remains an open question, but conceptually I measures orientation toward complexity. States with higher I values are those that better support:

- Structure formation and self-organization,
- Coherent dynamics and pattern emergence,
- Complexity growth and hierarchical organization,
- Stable, reproducible behavior.

The information term $(1 + \alpha I)$ in the coupling function biases selection toward configurations with higher I values. This bias ensures that, among states with similar energies, those that better support complexity are preferentially selected.

4.1.3 The Coupling Mechanism

The coupling of energy and information through the product structure $f(E, I) = \exp[-(E - E_c)^2/(2\sigma^2)](1 + \alpha I)$ creates a *two-factor selection mechanism*:

1. Stability Gate: The complexity parameter $X = E \cdot I$ must fall within a viable range for stability. This creates a gate that filters out configurations that cannot support stable law formation.

2. Lock-in Trigger: The asymmetry parameter $|E - I|$ determines the likelihood of achieving rapid and decisive law lock-in. Large asymmetry breaks symmetry and propels the system toward a single, unchanging configuration.

Together, these factors explain why the most successful universes (those that stabilize their laws) represent an optimal compromise: they are energetic enough to initiate cosmogenesis, yet possess sufficient information orientation to guide the selection toward complexity-permitting laws.

4.1.4 Why Both Are Necessary

Energy alone is insufficient: fluctuations with the correct energy scale can occur, but without information orientation, the system lacks direction toward complexity. Information alone is also insufficient: orientation toward complexity requires sufficient energy to overcome thresholds and initiate stabilization.

The coupling ensures that only configurations with *both* the correct energy scale *and* sufficient information orientation can stabilize their laws. This two-factor requirement explains the apparent rarity of complexity-permitting universes: they must satisfy two independent conditions simultaneously.

4.2 Cosmogenesis Mechanism

TQE provides a mechanistic account of cosmogenesis—the origin of the universe—without requiring external agency or pre-existing laws. The mechanism operates entirely through internal dynamics in the pre-law phase.

4.2.1 Initial State

The process begins in a *pre-law state*—a regime devoid of classical space-time and established physical laws. In this state, the system is governed by a probability distribution $P(\psi)$ over the space Ψ of possible law-states. This distribution evolves according to the modulation rule (2), which depends on energy and information but not on space-time coordinates (which do not exist).

4.2.2 Fluctuation and Selection

A large energy fluctuation occurs, enabled by quantum uncertainty principles. This fluctuation provides sufficient energy E to initiate cosmogenesis. Simultaneously, information orientation I emerges (through mechanisms that remain to be fully understood; see Section 2.3).

The coupling function $f(E, I)$ evaluates each candidate law-state ψ based on its (E, I) properties. States with favorable combinations receive enhanced statistical weight, while others are suppressed. The modulation rule (2) updates the probability distribution, concentrating probability around law-compatible configurations.

4.2.3 Convergence and Lock-in

As the system evolves, configurations with favorable (E, I) combinations gain statistical weight. The probability distribution $P(\psi)$ begins to concentrate around law-compatible states. When the complexity parameter $X = E \cdot I$ enters the viable parameter range and the asymmetry $|E - I|$ exceeds a threshold, the system crosses a critical point.

At this critical point, the probability distribution rapidly concentrates on a narrow subset of configurations. The selected configuration exhibits persistent stability, satisfying the lock-in criterion (19). Law lock-in is achieved:

a quasi-stationary set of physical laws stabilizes and becomes effectively permanent.

4.2.4 Emergence of Classical Regime

Once law lock-in occurs, classical space-time emerges. The selected law-state ψ includes a metric structure $g_{\mu\nu}$ (encoded in ψ), allowing classical space-time to emerge. General Relativity becomes applicable, and the universe evolves according to the stabilized laws.

The transition from pre-law to law-governed regime marks the emergence of classical physics from the probabilistic selection process. Space-time itself is emergent—it arises from law selection, not as a pre-existing structure.

4.2.5 Why No New Universes Form

Once laws are locked in, the stabilized laws suppress large fluctuations that could seed new universes within the existing one. The coupling function $f(E, I)$, now operating with fixed laws, exponentially suppresses configurations that violate the established laws. This explains why new universes do not form within an existing universe: the laws themselves act as a suppression mechanism.

4.3 Relation to Quantum Mechanics and Gravity

TQE provides a framework that connects to both quantum mechanics and general relativity, though the precise mathematical connections remain open questions.

4.3.1 Quantum Mechanics

The probabilistic nature of TQE—its use of probability distributions $P(\psi)$ and probabilistic selection—has conceptual parallels with quantum mechanics. Both frameworks employ probability to describe fundamental uncertainty, and both involve the evolution of probability distributions over time.

In quantum mechanics, the wave function $\Psi(x, t)$ evolves according to the Schrödinger equation, and probabilities are computed via the Born rule $P(x) = |\Psi(x)|^2$. In TQE, the probability distribution $P(\psi)$ evolves according to the modulation rule (2), which depends on energy and information.

The quantum mechanical collapse mechanism (wave function collapse upon measurement) has a parallel in TQE’s lock-in mechanism (probabilistic concentration upon law selection). Both processes involve a transition from a probabilistic superposition to a definite outcome.

Open Question 4.1 (Quantum Foundation). Can TQE’s probabilistic structure be derived from quantum mechanics? How does the modulation rule (2) relate to quantum evolution and measurement? Can the information parameter I be defined in terms of quantum information measures (e.g., entanglement entropy, quantum Fisher information)?

4.3.2 General Relativity

TQE provides a mechanism for the *emergence* of space-time from law selection. The pre-law phase has no classical space-time structure, but after law lock-in, space-time emerges as part of the selected law-state ψ . This emergence provides a natural explanation for why space-time exists and why it has the properties it does: space-time is selected through the probabilistic mechanism, and its properties are encoded in ψ .

The relationship between TQE and GR involves a conceptual reversal: rather than space-time existing a priori and constraining the evolution of matter (as in GR), TQE proposes that space-time emerges from the selection of law-compatible configurations.

Remark 4.1. This perspective aligns with approaches in quantum gravity that treat space-time as emergent (e.g., loop quantum gravity, causal dynamical triangulation, string theory). However, TQE provides a different mechanism for space-time emergence through law selection rather than geometric or field-theoretic dynamics.

The Hawking-Penrose singularity theorems [6, 7] do not apply in the pre-law phase because their prerequisites (classical space-time structure, energy conditions) are not satisfied. This observation reframes the singularity problem: rather than accepting an initial singularity as inevitable, TQE proposes that the pre-law phase lacks the structures that make singularities meaningful.

Open Question 4.2 (GR Connection). How does the metric structure emerge from law selection? What is the precise mathematical relationship between the selected law-state ψ and the emergent metric $g_{\mu\nu}$? Can the Einstein field equations be derived from TQE’s selection mechanism?

4.3.3 Unification Perspective

TQE offers a unified perspective where both quantum mechanics and general relativity emerge from a deeper probabilistic selection process. The quantum mechanical probabilistic structure and the gravitational space-time structure both arise from the modulation rule (2) operating in the pre-law phase.

This unification differs from standard approaches (e.g., string theory, loop quantum gravity) that attempt to quantize gravity or unify forces at high energies. Instead, TQE proposes that both quantum mechanics and gravity emerge from the same selection process that selects physical laws.

Open Question 4.3 (Unification). Can TQE provide a unified framework for quantum mechanics and general relativity? How do quantum effects and gravitational effects emerge from the same selection mechanism? What are the implications for quantum gravity?

These questions represent some of the most important open directions for TQE’s development, as they would establish direct connections between the framework and established physics.

5 Discussion and Outlook

This section discusses TQE’s place in the broader landscape of theoretical physics, its relationship to other approaches, and open questions for future development.

5.1 Falsifiable Predictions

A crucial aspect of TQE is that it makes falsifiable predictions, though these are currently at a theoretical level. Quantitative predictions require further development and comparison with observations.

5.1.1 Theoretical Predictions

If TQE is correct, one expects:

1. Non-random Statistical Features: The selection mechanism operating in the pre-law phase should imprint non-random statistical features on the universe. These features should be observable in cosmological observations, particularly in large-scale anomalies in the cosmic microwave background (CMB).

2. Statistical Patterns: The stabilization mechanism should create statistical patterns that distinguish the universe from random realizations. These patterns may manifest as:

- Anomalous alignments in CMB multipoles (e.g., “Axis of Evil”),
- Non-random distribution of large-scale structures,
- Statistical signatures of law stabilization in the early universe.

3. Diagnostic Proxies: Specific cosmological anomalies (e.g., low quadrupole, hemispherical power asymmetry) may serve as diagnostic proxies for the stabilization dynamics. These anomalies should not be random but should exhibit patterns consistent with the selection mechanism.

5.1.2 Theoretical Level Only

I emphasize that these are *theoretical predictions* at this stage. Quantitative predictions (e.g., specific angular scales, statistical significances) require:

- Detailed numerical simulations of the selection mechanism,
- Comparison with observational data (e.g., Planck CMB maps),
- Statistical analysis to test for predicted patterns.

These quantitative predictions represent a separate phase of research that builds upon the theoretical framework presented here.

Open Question 5.1 (Quantitative Predictions). Can TQE make quantitative, testable predictions for specific cosmological observables? What are the precise statistical signatures of the selection mechanism, and how can they be distinguished from random fluctuations?

5.2 Relation to Other Theories

TQE shares conceptual connections with several other approaches in theoretical physics, while also differing in important ways.

5.2.1 Wheeler’s “It from Bit”

Wheeler’s proposal that “information is fundamental” [13] has conceptual parallels with TQE’s information parameter I . However, TQE provides a *quantitative framework* with specific mathematical structure (the modulation rule (2) and coupling function (5)) rather than a purely conceptual statement.

Wheeler’s framework emphasizes that information may be more fundamental than matter-energy, but does not provide a specific mechanism for how information influences physical laws. TQE proposes a concrete mechanism through the energy-information coupling.

5.2.2 Zurek’s Quantum Darwinism

Zurek’s quantum Darwinism [14] explores how certain quantum states are selected for their robustness and information-carrying capacity. This selection process has parallels with TQE’s selection of law-compatible configurations.

However, quantum Darwinism focuses on the selection of quantum states through interaction with the environment (decoherence), while TQE focuses on the selection of physical laws through energy-information coupling. The selection mechanisms differ, though both involve probabilistic selection based on information-theoretic criteria.

5.2.3 Tegmark’s Mathematical Universe Hypothesis

Tegmark’s Mathematical Universe Hypothesis [11] posits that mathematical structures exist independently and physical reality corresponds to specific mathematical patterns. This has conceptual parallels with TQE’s space Ψ of possible law-states.

However, Tegmark’s framework treats all mathematical structures as equally real, while TQE proposes a *selection mechanism* that preferentially weights certain configurations. TQE provides a mechanism for *why* one mathematical structure is realized rather than another, while Tegmark’s framework lacks such a mechanism.

5.2.4 Davies on Emergent Laws

Davies has argued for the emergence of laws from deeper principles [4], proposing that physical laws may not be fundamental but emerge from more primitive dynamics. This perspective aligns with TQE’s central hypothesis.

However, Davies’ framework does not provide a specific mechanism for law emergence, while TQE proposes a concrete probabilistic selection mechanism through energy-information coupling.

5.2.5 Smolin’s Cosmological Natural Selection

Smolin’s cosmological natural selection [9] proposes that universes evolve through black hole production, selecting laws that favor black hole formation. This framework provides a mechanism for law selection through cosmological evolution.

TQE differs in that it proposes selection during cosmogenesis (pre-law phase) rather than through cosmological evolution (post-law phase). The selection mechanisms differ: Smolin’s framework uses black hole production

as a selection criterion, while TQE uses energy-information coupling during the initial phases of the universe.

5.2.6 Key Differences

What distinguishes TQE from these approaches is:

1. **Quantitative Framework:** TQE provides specific mathematical structure (modulation rule, coupling function) rather than purely conceptual statements.
2. **Falsifiable Mechanism:** The framework yields falsifiable predictions through its selection mechanism, making it testable in principle.
3. **Pre-law Selection:** TQE proposes selection during cosmogenesis (pre-law phase), before classical space-time exists, rather than through cosmological evolution.
4. **Energy-Information Coupling:** The specific coupling mechanism through $f(E, I)$ provides a concrete implementation of law selection.

5.3 Future Directions

TQE is a framework in development, and many questions remain open. This section outlines key directions for future research.

5.3.1 Fundamental Questions

Open Question 5.2 (Information Parameter Definition). What is the fundamental definition of the information parameter I ? Can it be derived from quantum field theory, quantum information theory, or holographic principles? Is it related to entanglement entropy, quantum Fisher information, or other information-theoretic quantities?

Open Question 5.3 (Pre-law to Law Transition). What is the precise mechanism by which the pre-law phase transitions to a law-governed regime? How does classical space-time emerge from law selection? What are the mathematical details of this transition?

Open Question 5.4 (Connection to Standard Physics). Can the laws of standard physics (conservation laws, thermodynamics, gravity, quantum mechanics) be rigorously derived from TQE's master coupling $f(E, I)$ through appropriate coarse-graining procedures? What are the precise mathematical steps?

5.3.2 Theoretical Development

Future theoretical work should focus on:

1. Rigorous Derivation: Developing rigorous derivations of standard physics from TQE’s master coupling, establishing the hierarchical mapping from Tier 0 to higher tiers.

2. Information Parameter Foundation: Establishing a fundamental theoretical foundation for the information parameter I , linking it to quantum information measures, entanglement entropy, or holographic principles.

3. Transition Mechanism: Developing a detailed mathematical description of the pre-law to law transition, including the emergence of classical space-time and the lock-in mechanism.

4. Stability Analysis: Performing rigorous stability analysis of fixed points, exploring the parameter space for viable universes, and characterizing the stability windows.

5.3.3 Quantitative Predictions

Future quantitative work should focus on:

1. Numerical Simulations: Developing numerical simulations of the selection mechanism to compute quantitative predictions for cosmological observables.

2. Statistical Analysis: Performing statistical analysis of simulated universes to identify patterns and signatures of the selection mechanism.

3. Observational Comparison: Comparing theoretical predictions with observational data (e.g., Planck CMB maps) to test the framework.

4. Parameter Estimation: Estimating the parameters (E_c, σ, α) from observations or theoretical constraints.

5.3.4 Experimental Tests

The development of experimental tests for TQE represents a crucial long-term goal:

Open Question 5.5 (Experimental Tests). How can TQE be tested experimentally? What are the observational signatures of the selection mechanism, and how can they be distinguished from other effects? What future observations or experiments could test the framework?

These questions represent some of the most important directions for TQE’s development, as they would establish the framework’s testability and connection to observations.

6 Conclusions

In this manuscript, I have presented the Theory of the Question of Existence (TQE), a theoretical framework where stable physical laws emerge dynamically from the coupling of vacuum energy fluctuations with an information-theoretic orientation parameter during cosmogenesis. The framework addresses one of the most profound questions in cosmology: why do stable, complexity-permitting physical laws exist at all?

6.1 Main Results

The main contributions of this work are:

1. **Core Framework:** TQE provides a quantitative framework for law emergence through the modulation rule (2), where probability distributions over law-states evolve according to the coupling function $f(E, I)$ that depends on energy and information.
2. **Pre-law Phase:** The framework postulates a pre-law phase devoid of classical space-time, where quantum fluctuations in the vacuum can initiate cosmogenesis without requiring external agency or pre-existing laws.
3. **Selection Mechanism:** The coupling of energy and information creates a two-factor selection mechanism (stability gate via $X = E \cdot I$, lock-in trigger via $|E - I|$) that preferentially weights law-compatible configurations.
4. **Law Lock-in:** The framework provides a mechanism for law stabilization through a lock-in process where selected configurations become quasi-stationary and effectively permanent.
5. **Emergent Space-time:** Classical space-time emerges from law selection, providing a mechanism for the origin of space-time structure without requiring a pre-existing metric.

6.2 Key Insights

The framework offers several key insights:

- **Fine-tuning Reframed:** The fine-tuning problem is reframed from “why are laws fine-tuned?” to “how do fine-tuned laws emerge dynamically?” TQE provides a mechanistic account without invoking anthropic filtering or infinite universes.

- **Laws as Emergent:** Physical laws are not pre-existing axioms but emerge from a selection process operating in the pre-law phase. This provides a natural explanation for why particular laws exist rather than alternatives.
- **Energy-Information Coupling:** The coupling of energy and information is proposed as a fundamental mechanism underlying law emergence. This coupling creates selection biases that steer the system toward complexity-permitting configurations.
- **Pre-law Regime:** The pre-law phase, devoid of classical space-time, provides a regime where selection can occur without the constraints of established physics. The Hawking-Penrose singularity theorems do not apply because their prerequisites are not satisfied.

6.3 Open Questions

Many questions remain open and require further development:

- **Information Parameter:** The fundamental definition of the information parameter I remains an open question, requiring theoretical development linking it to quantum information measures or holographic principles.
- **Connection to Standard Physics:** The rigorous derivation of standard physics from TQE's master coupling remains to be established, requiring development of the hierarchical mapping from Tier 0 to higher tiers.
- **Quantitative Predictions:** While the framework makes theoretical predictions, quantitative predictions require numerical simulations and comparison with observations.
- **Experimental Tests:** The development of specific experimental tests for TQE remains a crucial goal for the framework's validation.

6.4 Future Work

Future work should focus on:

1. Establishing a fundamental theoretical foundation for the information parameter I .

2. Developing rigorous derivations of standard physics from TQE’s master coupling.
3. Performing numerical simulations to compute quantitative predictions.
4. Comparing theoretical predictions with observational data.
5. Developing specific experimental tests for the framework.

6.5 Final Remarks

TQE offers a novel, mechanism-based answer to the question of why stable physical laws exist. Rather than invoking anthropic reasoning or infinite universes, the framework provides a quantitative, falsifiable mechanism for law emergence through energy-information coupling during cosmogenesis.

While many questions remain open, the framework provides a promising direction for understanding the origin of physical laws and the fine-tuning problem. The development of TQE represents an ongoing research program that requires continued theoretical development, numerical simulations, and comparison with observations.

The framework’s testability through falsifiable predictions provides a path forward: as theoretical predictions are developed and compared with observations, TQE can be refined, tested, and potentially validated or falsified. This testability is crucial for moving the question of law origin from philosophical speculation to quantitative, falsifiable physics.

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